codecs only extends to 3.4kHz. Any frequencies above that limit are simply cut off, which is why phone calls sound so muffled. The new codec allows frequencies of up to 16 or even 20kHz to be transmitted, depending on the bit rate of the connection."

Computing system inspired by human brain

A team of researchers from Georgia Institute of Technology and University of Notre Dame has created a computing system that employs a network of electronic oscillators to solve graph colouring tasks—a problem that tends to choke today's computers.

Graph colouring problem starts with a graph—a visual representation of a set of objects connected in some way. To solve the problem, each object must be assigned a colour, but two objects directly connected cannot share the same colour. Typically, the goal is to colour all objects in the graph using the smallest number of different colours.

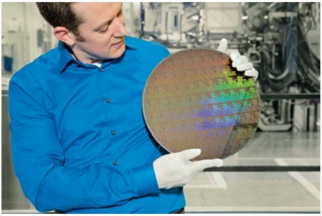
In designing a system different from traditional transistor-based computing, the researchers took cues from the human brain, where processing is handled collectively, such as a neural oscillatory network, rather than with a central processor.

The electronic oscillators, fabricated from vanadium dioxide, were found to have a natural ability that could be harnessed for graph colouring problems. When a group of oscillators were electrically connected via capacitive links, these automatically synchronised to the same frequency—oscillating at the same rate. Meanwhile, oscillators directly connected to one another would operate at different phases within the same frequency, and oscillators in the same group but not directly connected would sync in both frequency and phase.

"If you suppose that each phase represents a different colour, this system was essentially mimicking naturally the solution to a graph colouring problem," said Arijit Raychowdhury, an associate professor in Georgia Tech's School of Electrical and Computer Engineering.

A groundbreaking 5-nanometre chip

A fingernail-sized chip housing a whopping 30 billion transistors, the on-off switches of electronic devices, could be a possibility soon. The impetus comes from development of transistors for a 5-nanometre semiconductor chip, jointly by IBM Research and partners GlobalFoundries and Samsung. Right now, the most advanced semiconductor chips use a FinFET process with circuitry that is 10 nanometres in width. Companies such as Intel can build chips with 10 or 15 billion transistors using that process.



IBM Research scientist Nicolas Loubet holds a wafer of chips with 5nm silicon nanosheet transistors (Image courtesy: http://www-03.ibm.com)

"A 5-nanometre chip could perform about 40 per cent faster than a 10-nanometre chip, given the same power settings," said Mukesh Khare, vice president of semiconductor technology research at IBM Research, in an interview with VentureBeat. Or a chip could be 75 per cent more power efficient.

IBM said the resulting increase in performance will accelerate cognitive computing, the Internet of Things (IoT), and other data-intensive applications delivered in the cloud. The power savings could also mean that the batteries in smartphones and other mobile products could last two to three times longer than today's devices before needing to be charged.

Scientists achieved the breakthrough by using stacks of silicon nanosheets for the transistor, instead of using the standard FinFET architecture. The transistors are built with extreme ultraviolet (EUV) lithography, which is used to stencil the circuit designs on chips. With EUV lithography, the width of the nanosheets can be adjusted continuously, all within a single manufacturing process or chip design.

New system to restore communications during a disaster

A portable system that allows communications to be restored in the wake of a disaster and helps direct survivors to safety is being devised by academia.

Ron Austin, associate professor of networks and security at Birmingham City University (UK), has created the prototype system, which could be used to plug a crucial gap in systems such as telephone, GPS and Internet links during the first 24 hours following a disaster.

The network runs using Raspberry Pi computer development boards, which can be linked together to form a bespoke setup tailored to the needs of a site. It